WORK SHEET - 09

Single Correct

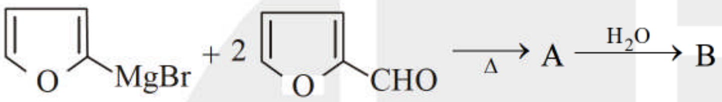
Q.1 Compound A (molecular formula C_3H_8O) is treated with acidified potassium dichromate to form a product B (molecular formula C_3H_6O). B forms a shining silver mirror on warming with ammoniacal silver nitrate, B when treated with an aqueous solution of $NH_2NHCONH_2$ and sodium acetae gives a product C. Identify the structure of C.

(A) $CH_3CH_2CH = NNHCONH_2$

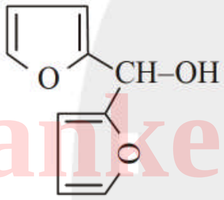
(B) $CH_3C(CH_3) = NHHCONH_2$

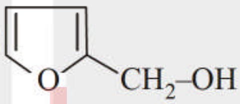
(C) $CH_3C(CH_3) = NCONHNH_2$

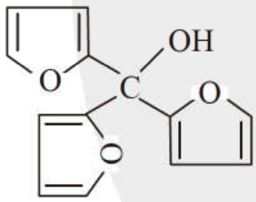
(D) $CH_3CH_2CH = NCONHNH_2$

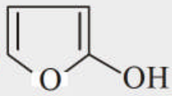
Q.2 

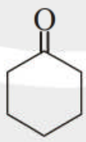
B is -

(A) 

(B) 

(C) 

(D) 

Q.3 (P) $\xrightarrow{HCN}$ (Q) $\xrightarrow{LiAlH_4}$ (R) $\xrightarrow{NaNO_2+HCl}$  (S)

Relationship between (P) & (S).

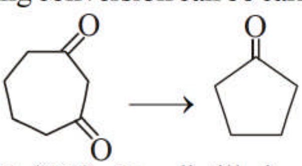
(A) Identical compound

(B) Homologues compound

(C) Position isomers

(D) Chain isomers

Q.4 Following conversion can be carried out by.



(A) I_2 / $Ca(OH)_2$, Dry distillation

(B) (i) I_2 / (ii) NaOH, CaO, Δ

(C) Δ

(D) NaOH, Δ

Q.5 $Cl-\underset{\underset{O}{\|}}{C}-\underset{\underset{O}{\|}}{C}-H \xrightarrow[\Delta]{NaOH(excess)}$

(A) $H-\overset{\overset{O}{\|}}{C}-COONa$

(B) $\underset{COONa}{\overset{COONa}{|}} + \underset{COONa}{\overset{CH_2OH}{|}}$

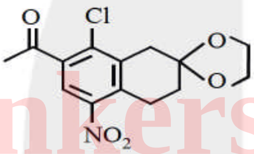
(C) $Cl-\underset{\underset{O}{\|}}{C}-COONa + Cl-\underset{\underset{O}{\|}}{C}-CH_2OH$

(D) $H-\overset{\overset{Cl}{|}}{\underset{\underset{OH}{|}}{C}}-COONa$

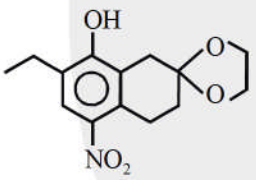
Q.6 $H_3C-\underset{\underset{CN}{|}}{CH}-\underset{\underset{O}{\|}}{C}-CH_3 \xrightarrow[(ii)\Delta]{(i)H^+/H_2O} P$

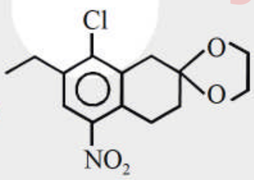
(A) P gives +ve iodoform test & –ve test with Fehling solution.
(B) P gives –ve iodoform test & +ve test with $NaHCO_3$ solution.
(C) P gives +ve Lucas test & –ve test with $NaHSO_3$ solution
(D) P gives +ve test with $NaHSO_3$ & cerric ammonium nitrate solution.

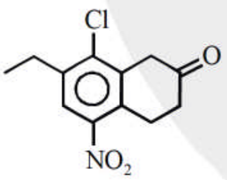
Q.7 Predict the major product of reaction :

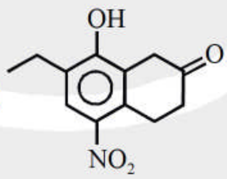


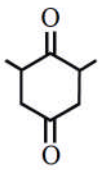
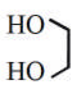
$\xrightarrow{N_2H_4/KOH}$ (A) ; Product (A) is :

(A) 

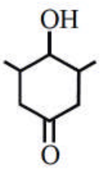
(B) 

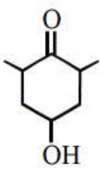
(C) 

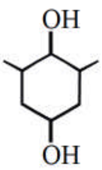
(D) 

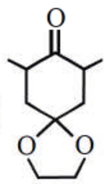
Q.8  +  $\xrightarrow{H^\oplus}$ A $\xrightarrow[2.\ H_3^\oplus O]{1.\ LiAlH_4}$ B

Identify structure of B :

(A) 

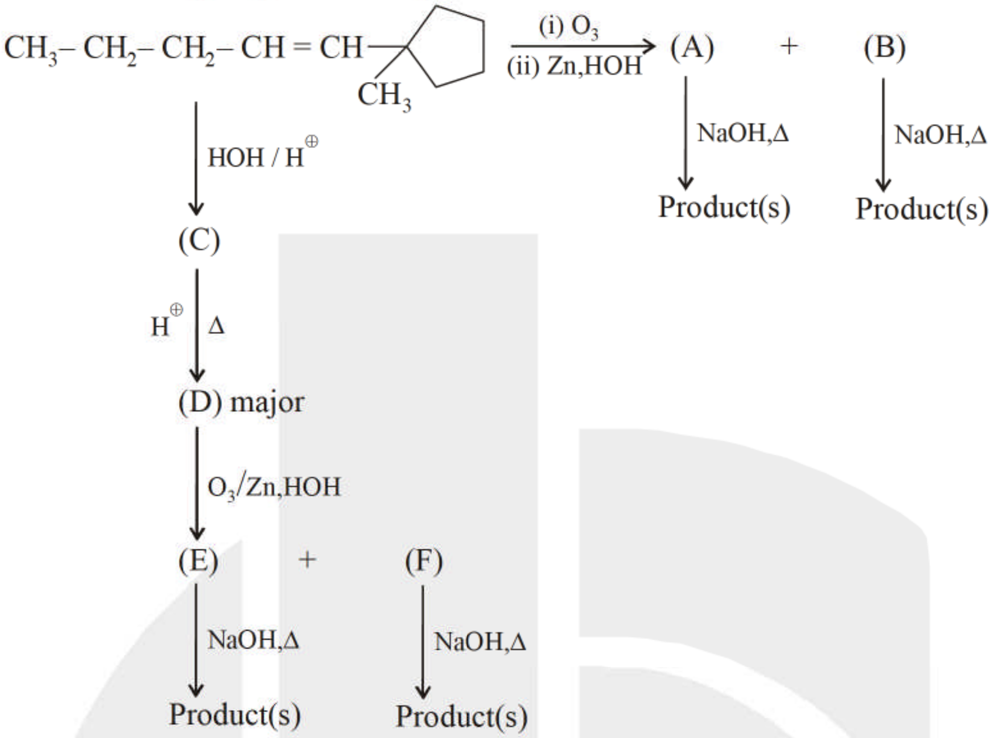
(B) 

(C) 

(D) 

Comprehension :

Paragraph for question nos. 9 to 11

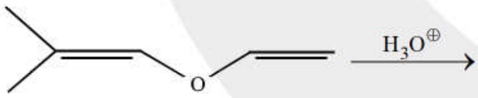


Q.9 Which of the following reaction or reaction mechanism is not involved in above interconversions?
(A) Electrophilic addition reaction (B) Acid base reaction
(C) Elimination reaction (D) Free radical combination reaction

Q.10 Number of ozonide formed during the formation of A & B (Excluding stereo)
(A) 1 (B) 2 (C) 3 (D) 4

Q.11 Find out the correct statement.
(A) In presence of NaOH, A and B give the same type of reaction
(B) In presence of NaOH, E and F give the same type of reaction
(C) D is same as starting compound
(D) Ring size is different in D and starting compound

More than one correct :

Q.12 

(A) + (B) formed cannot be differentiated by
(A) Iodoform (B) Fehling (C) $NaHSO_3$ (D) 2,4-DNP

Q.13 Which of the following option(s) are correct:

(A) $CH_3-\overset{\overset{\large O}{||}}{C}-CH_2CH_3 \xrightarrow[\text{NaOH (1eq.)}]{Br_2\text{ (1eq.)}} \underset{\text{(Major product)}}{Br-CH_2-\overset{\overset{\large O}{||}}{C}-CH_2CH_3}$

(B) $CH_2(SO_3R)_2$ can be use as a reactant for Knoevenagel reaction

(C) Tischenko reaction is catalysed by $(C_2H_5O)_3Al$.
(D) Intramolecular clasien condensation is known as Dieckmann's condensation.

Q.14 Which of the following do not give Cannizzaro reaction ?

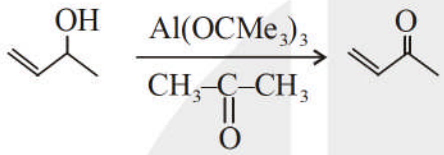
(A) $Cl_3C-\overset{\overset{O}{||}}{C}-H$ (B) $H-\underset{\underset{O}{||}}{C}-H$ (C) $Ph-\underset{\underset{O}{||}}{C}-H$ (D) $Me_2\ddot{N}-C_6H_4-\overset{\overset{O}{||}}{C}-H$

Q.15 Which of the following reactions involve hydride ion transfer ?

(A) $2H-\overset{\overset{O}{||}}{C}-H \xrightarrow{\overset{\ominus}{O}H} H-\overset{\overset{O}{||}}{C}-\overset{\ominus}{O} + CH_3-OH$

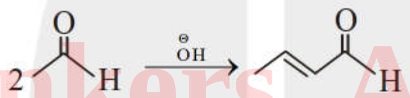
(B) $2H_3C-\overset{\overset{O}{||}}{C}-H \xrightarrow[\Delta]{\overset{\ominus}{O}H} H_3C-CH=CH-\overset{\overset{O}{||}}{C}-H$

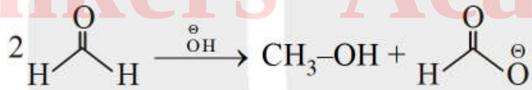
(C) $2H_3C-\overset{\overset{O}{||}}{C}-H \xrightarrow{Al(OEt)_3} CH_3-\overset{\overset{O}{||}}{C}-O-CH_2CH_3$

(D) 

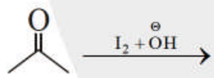
Match the column :

Q.16 **Column (I)** **Column (II)**

(A)  P Oxidation

(B)  Q. Condensation

(C) $2Ph-\overset{\overset{O}{||}}{C}-H \xrightarrow{KCN}$ Product R. Nucleophilic addition

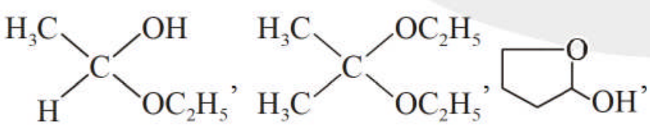
(D)  Product S. Electrophilic substitution

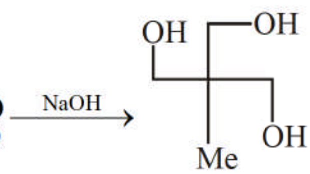
T. Nucleophilic substitution

Subjective :

Q.17 Of the following compounds, how many would give positive silver mirror test.

$CH_3-\overset{\overset{O}{||}}{C}-H$, $CH_3-\overset{\overset{O}{||}}{C}-OH$, $H-\overset{\overset{O}{||}}{C}-OH$, $CH_3-\overset{\overset{O}{||}}{C}-CH_3$, $Ph-\overset{\overset{O}{||}}{C}-H$

, $CH_3-C\equiv C-H$

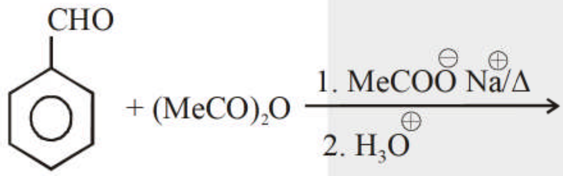
Q.18 $MeCH_2CHO + \underset{(excess)}{HCHO}$ 

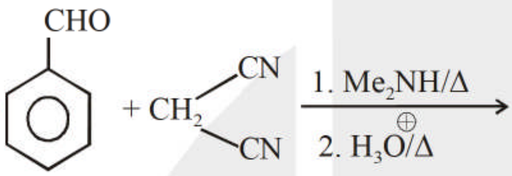
How many aldol reactions will take place in above conversion :

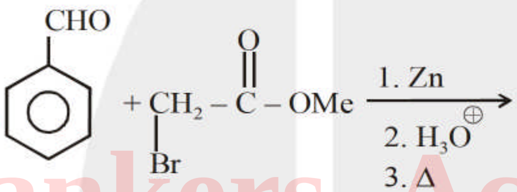
Q.19 Which of the following test is/are given by $\underset{\displaystyle OH}{\underset{|}{CH_2}} - \overset{\displaystyle O}{\overset{\|}{C}} - CH_2 - CH_2 - CH_2 - \underset{\displaystyle OH}{\underset{|}{CH_2}}$ the compound?

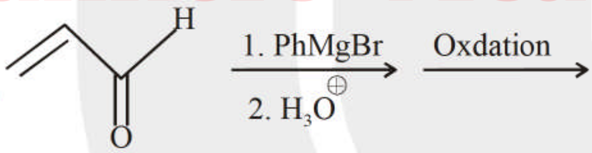
2,4-DNP reagent, Tollen's test, Fehling test, Schiff's reagent, $NaHSO_3$, Cerric ammonium nitrate, aq. $FeCl_3$, Litmus Test, Haloform

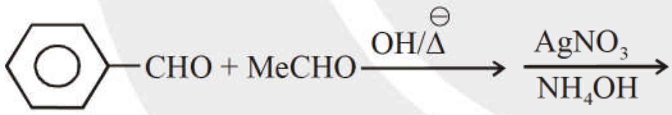
Q.20 Which reation will give cinnamic acid as final product?

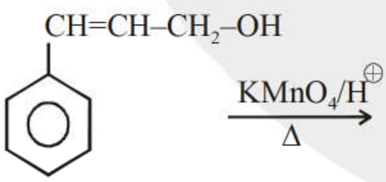
(1) 

(2) 

(3) 

(4) 

(5) 

(6) 

ANSWER KEY

Q.1	A	Q.2	B	Q.3	B	Q.4	A	Q.5	B
Q.6	A	Q.7	A	Q.8	A	Q.9	D	Q.10	C
Q.11	B	Q.12	BCD	Q.13	ABCD	Q.14	AD	Q.15	ACD
Q.16	[(A)–Q, R (B)–PRT, (C) – QR, (D)–P,S,T]					Q.17	[5]	Q.18	[2]
Q.19	[5]	Q.20	[4]						